

Residual Minimization And Scientific ML Reliability — cycles 10-12

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Residual Minimization And Scientific ML Reliability — cycles 10-12

Abstract

Cycles 10-12 were closure and handoff cycles. They did not add new mathematical results, code, figures, tests, or scientific claims. Instead, each cycle confirmed that the previously validated package already satisfied the directive: it includes a broad residual-failure catalogue, an application map, proof and obstruction notes, toy simulations, a synthesis package, a final reader-facing report, tests, validator checks, and ledger validation.

The repeated decision across cycles 10, 11, and 12 was **PIVOT**. In this context, PIVOT does not mean the research package failed. It means a closure-only cycle cannot be newly validated as substantive research under the no-null-cycle rule because it produces no new evidence. The process-level conclusion is to stop research/audit cycling and hand the package to the user.

The supported thesis remains unchanged: residual objectives certify physical behavior only when they are coercive or correctly matched to the claimed physical behavior. The package preserves its limitations and does not claim that physics-informed neural networks, neural operators, collocation methods, or residual minimization methods fail in general [1]-[5].

Introduction

This project investigated when residual minimization can produce small loss while failing to certify the desired physical behavior in scientific machine learning. A residual loss measures how well a candidate function satisfies a differential equation, constraint, or learned operator objective. A certificate is an additional norm, condition, estimate, or constraint that makes the residual objective imply the physical property being claimed.

Earlier cycles built the substantive research package: a catalogue of residual-minimization mechanisms, application mapping, representative toy simulations, branch reports, and final synthesis. Cycles 10-12 did not reopen any of that scientific work. Their purpose was to

confirm whether any blocker remained after the final delivery package had already been validated.

The answer across all three cycles was no. The only remaining loose ends are labeled limitations rather than blockers: standalone M-2, standalone M-4, CAT-18, and CAT-20.

Approach

The report uses the provided cycle-session records and audit report as source material. The cycles were closure-only, so the structure is chronological rather than mechanism-by-mechanism.

Cycle	Role	Session ID	Source content
10	Researcher	e2ea5d07-6e54-4445-a78e-ddbc777729ed	Closure brief: no open research questions unless M-2, M-4, CAT-18, or CAT-20 is explicitly reopened.
10	Worker	65523af7-fff5-44c2-a98e-1cd0f5c09d32	No new artifact or command; final package already validated.
10	Auditor	78bba073-7f70-400c-b557-75bb0d9d6348	PIVOT decision; validators pass with historical warnings only.
11	Researcher	43cf4199-c349-4fd1-8b91-282a2a643fd0	Closure confirmation; no build, run, audit, or fanout recommended.
11	Worker	b3d1b457-da03-4054-85ec-081b1656478e	No new artifact or run; final package remains ready for use.
11	Auditor	22636c13-2e81-484a-95de-e5891cb6bdb8	PIVOT decision; no new artifact, build, branch, or blocker.
12	Researcher	a0f47480-6758-4361-ac5f-b3ae4ce03c5c	Closure and handoff brief; breadth criteria already satisfied.
12	Worker	5a26fdd7-6ff0-4d22-95b0-d2de4913b6e8	No commands run; final package ready for human use.
12	Auditor	bf62ea32-f87b-44cf-9b79-bc1ed9b18c9c	PIVOT decision; no new evidence, command output, test result, or scientific claim.

Findings

Cycle 10: First Closure Confirmation

Cycle 10 confirmed that the validated final-delivery state was already sufficient. The researcher session identified the current sub-topic as session closure and stated that no open research questions remained unless the user explicitly reopened standalone M-2, standalone M-4, CAT-18, or CAT-20.

The worker built nothing new and ran no commands. The worker record pointed back to the already validated reproduction state: the focused toy suite had passed with 35 passed, `promise_check` had passed with historical warnings only, and `org_check` had passed with historical warnings only.

The auditor issued PIVOT. The rationale was procedural: a closure-only cycle with no new substantive result, build, test, or artifact cannot receive a fresh VALIDATED judgment. The auditor also confirmed that the final report existed at `reports/residual_minimization_reliability_final_report.md`, that `_run/final-delivery-ready` remained validated, and that the only warnings were historical or organizational.

Cycle 11: Repeated No-Blocker Closure

Cycle 11 repeated the same closure state. The researcher record stated that final delivery was already validated, required breadth was met, and remaining warnings were historical or organizational.

The worker again built nothing and ran no commands. The worker listed the final package components that remained ready for human use:

- `reports/residual_minimization_reliability_final_report.md`
- `residual-certificates/broad_synthesis_package.md`
- `residual-certificates/residual_case_catalogue.md`
- `residual-certificates/application_risk_map.md`
- validated branch notes, scripts, tests, CSVs, and PNGs

The auditor again issued PIVOT. The decision was not based on a defect in the research package. It was based on the fact that cycle 11 was another closure-confirmation/null cycle with no new substantive research, build, test, or artifact. The appropriate action remained to stop cycling unless the user reopened M-2, M-4, CAT-18, CAT-20, or supplied a new audit blocker.

Cycle 12: Final Handoff Guidance

Cycle 12 made the closure instruction explicit. The researcher session identified the sub-topic as closure and handoff, with no new build, run, fanout, or artifact recommended. It also recorded the validated topic map: M-1, M-3, M-5, M-6, M-7, M-8, M-9, M-10, M-11, M-12, and `_run/final-delivery-ready` were validated; M-2 and M-4 remained pending or partly absorbed; CAT-18 and CAT-20 remained deferred.

The worker again built nothing and ran no commands. The worker stated that prior audited validation already covered the final report, catalogue, application map, branch reports, toy simulations, tests, `promise_check`, `org_check`, and ledger state.

The auditor issued PIVOT for the third time. The supplied audit report matches the session record: there were no critical or moderate findings, only documented historical warnings. The auditor confirmed that no new artifact, command output, test result, or scientific claim was produced in the cycle. The correct pivot is process-level handoff to the user.

Discussion

Cycles 10-12 show process closure, not new scientific content. The substantive package had already met the directive through earlier validated outputs: the catalogue, application map, proof and obstruction notes, toy simulations, broad synthesis package, and final reader-facing report.

The repeated PIVOT decisions prevent a closure loop from being mistaken for additional validation. They preserve the distinction between two different facts:

- The research package is complete for current purposes.
- A null cycle that adds no evidence should not be labeled as a new validated research result.

The package's scientific thesis remains narrow. Residual objectives can fail as certificates when they do not control the relevant physical behavior, for example because of nullspaces, weak topology, trace leakage, hidden state projection, missing admissibility constraints, underintegration, or deployment-region mismatch. Conversely, matched objectives, coercive norms, admissibility checks, energy estimates, entropy conditions, observability conditions, and related certificates can restore reliability in the studied settings.

Open Questions

The open questions are unchanged from the validated final package:

- **M-2:** A standalone continuous norm-mismatch theorem remains pending or partly absorbed by weak-topology and trace-leakage work.
- **M-4:** A standalone conservation-law selection campaign remains partly absorbed by the Burgers admissibility note; no full shock-selection simulation campaign was completed.
- **CAT-18:** The inverse PDE/source sensor-nullspace case remains deferred.
- **CAT-20:** The long-horizon surrogate drift case remains deferred.

Cycles 10-12 did not attempt to resolve these items. The source records state that they should be reopened only by explicit user request or a new blocker.

References

- [1] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, 378, 686-707, 2019. <https://doi.org/10.1016/j.jcp.2018.10.045>
- [2] P. B. Bochev and M. D. Gunzburger, "Least-Squares Finite Element Methods," *Applied Mathematical Sciences* 166, Springer, 2009. <https://link.springer.com/book/10.1007/b13382>
- [3] A. Quarteroni and A. Valli, "Numerical Approximation of Partial Differential Equations," *Springer Series in Computational Mathematics* 23, Springer, 1994. <https://link.springer.com/book/10.1007/978-3-540-85268-1>
- [4] H. Wendland, "Scattered Data Approximation," *Cambridge Monographs on Applied and Computational Mathematics* 17, Cambridge University Press, 2005. <https://www.cambridge.org/core/books/scattered-data-approximation/966D4683385F92FB8D1535F8D0A9585E>
- [5] R. J. LeVeque, "Finite Volume Methods for Hyperbolic Problems," *Cambridge Texts in Applied Mathematics*, Cambridge University Press, 2002. <https://www.cambridge.org/core/books/finite-volume-methods-for-hyperbolic-problems/CB7B0A27A6D37AE3B906D4AE7C60A708>

Appendix: Implementation Details

Code Organization

Cycles 10-12 produced no new code, data, figure, proof note, or scientific artifact.

The final reader-facing artifact remains:

- reports/residual_minimization_reliability_final_report.md

The main validated supporting artifacts remain:

- residual-certificates/broad_synthesis_package.md
- residual-certificates/residual_case_catalogue.md
- residual-certificates/application_risk_map.md
- residual-certificates/toy_simulation_results.md
- residual-certificates/weak_topology_branch.md
- residual-certificates/admissibility_invariant_branch.md
- residual-certificates/ode_reliability_branch.md
- residual-certificates/collocation_blind_spot_theorem.md

The executable toy suite remains organized under scripts/, tests/, and data/.

Test Results

No commands were run in cycles 10-12. The records refer back to the previously audited validation state:

- Focused toy suite: 35 passed.
- promise_check: exit 0, warnings only.
- org_check: exit 0, warnings only.
- _run/final-delivery-ready: validated in the ledger.

The historical warnings remain non-blocking: M-2 and M-4 are not standalone validated, CAT-18 and CAT-20 are deferred, old manager-assessment artifacts are missing, and some organizational warnings persist.

File Counts

MANIFEST.md was refreshed as a current workspace snapshot. The tracked counts are:

Group	Files	Lines
Scripts/check scripts	12	1341
Tests	11	554
Residual-certificate notes	13	998
Final/report/cycle-report markdown files	6	970
Root reference/ledger files	2	58
Total tracked files	44	3921

The report for cycles 10-12 itself is not included in those counts because the orchestrator writes it after this output block is consumed.

Session References

Cycle 10:

- Researcher: e2ea5d07-6e54-4445-a78e-ddbc777729ed
- Worker: 65523af7-fff5-44c2-a98e-1cd0f5c09d32
- Auditor: 78bba073-7f70-400c-b557-75bb0d9d6348

Cycle 11:

- Researcher: 43cf4199-c349-4fd1-8b91-282a2a643fd0
- Worker: b3d1b457-da03-4054-85ec-081b1656478e
- Auditor: 22636c13-2e81-484a-95de-e5891cb6bdb8

Cycle 12:

- Researcher: a0f47480-6758-4361-ac5f-b3ae4ce03c5c
- Worker: 5a26fdd7-6ff0-4d22-95b0-d2de4913b6e8
- Auditor: bf62ea32-f87b-44cf-9b79-bc1ed9b18c9c

Cross-Reference Map

- Final thesis -> reports/residual_minimization_reliability_final_report.md and residual-certificates/broad_synthesis_package.md
- Catalogue breadth -> residual-certificates/residual_case_catalogue.md
- Application mapping -> residual-certificates/application_risk_map.md
- Toy evidence -> residual-certificates/toy_simulation_results.md, scripts/, tests/, and data/
- Branch evidence -> residual-certificates/weak_topology_branch.md, residual-certificates/admissibility_invariant_branch.md, and residual-certificates/ode_reliability_branch.md
- Final validation ledger -> promise_ledger.jsonl
- Closure-cycle decisions -> cycle 10-12 researcher, worker, and auditor session records listed above